Workforce intelligence. With people in control.

A locally demonstrated recruitment and workforce intelligence platform. Private resume parsing and constrained DeepSeek-R1 evidence selection support accountable human reviews; they do not rank candidates, infer personal suitability or advance applications automatically.

VERIFIED LOCAL DEMONSTRATION

Real DeepSeek 1.5B + nomic-embed-text / Compose / 03 October 2026

01 / ARCHITECTURE AND AUTHORITIES

React / nginx
Scoped role workspaces

ASP.NET Core 10
Auth, jobs and domain rules

FastAPI / Ollama
Private evidence selection

PostgreSQL
Auditable state + pgvector

Data is authoritative in PostgreSQL

EF Core owns users/roles, departments, vacancies, applications, source documents, factual assertions, reviewer corrections, versioned analyses, interviews, manual decisions and audits. Model output is never the source of truth for a hiring status.

02 / SOURCE TO EVIDENCE PIPELINE

PDF / DOCX, not screenshots

Private 5 MiB uploads preserve original source text with page/paragraph IDs. No OCR or arbitrary resume-layout understanding is claimed.

Conservative fact capture

Explicit experience, education, project and skill mentions are extracted against a canonical taxonomy. Every new fact begins Unverified.

Reviewer-controlled truth

Recruiters can correct mappings, mark Documented/Ambiguous/Rejected and annotate changes. Original quotes and human edits stay distinguishable.

DeepSeek-R1 Distill 1.5B

Ollama executes the small Qwen-distilled model locally. Schema-constrained JSON selects requirement-to-evidence references rather than writing unrestricted hiring assessments.

Independent safety checks

Per-skill membership, coverage and omission checks reject unsupported reference combinations. Checked templates generate the displayed explanations.

Embedding retrieval, not scoring

nomic-embed-text produces 768-dimensional taxonomy embeddings; pgvector cosine/HNSW finds similar skill names, never candidate rankings.

DESIGN RULE Missing means undocumented. Similarity is not employability. The human reviewer chooses what to verify, what to pursue and what to decide.

Implemented. Tested. Bounded.

Verification comes from the repository's 3 October 2026 local handoff. This brief distinguishes a successful real-inference smoke run from browser coverage, configured CI and unexecuted provisioning.

03 / IMPLEMENTATION AND SECURITY

Backend: .NET 10 + EF Core

Durable processing jobs use PostgreSQL advisory locking, restart recovery, three attempts and visible failure states. Evidence edits enforce a revision token (HTTP 409).

Document service: Python / FastAPI

Signature and archive validation reject unsafe PDF/DOCX inputs. Private internal-key authentication; no published AI/doc-service port.

Identity and access

PBKDF2-SHA256 password hashes, two-hour JWTs in browser memory, server roles and ownership. Candidates cannot see recruiter-only analysis or notes.

Data minimisation

DeepSeek sees canonical requirement and evidence IDs/statuses; not resume contents, identities or protected traits. Proficiency is never inferred.

Retention and erasure

Original documents expire after 180 days; deletion cascades to source facts, analyses and review notes. Backup retention needs separate policy.

Local operations

Compose is the verified runtime; OpenTofu Docker-provider formatting/init/validation passed, but deployment was not applied. GitHub Actions workflow was committed, not run.

04 / EXECUTED EVIDENCE

10
Python safety checks

768
Embedding dimensions

3
Worker attempts

Passed: .NET and React builds; ten targeted Python checks; standard API smoke with explicit failed-model handling; real-model **--require-ai** run using DeepSeek-R1 and nomic-embed-text, checking persisted inference, pgvector retrieval, review corrections, stale analyses, role isolation, interview scheduling, workforce coverage, manual decisions and document/application erasure.

Observed but limited: HR/candidate sign-in, navigation and vacancy/department creation received browser checks. There was no full browser regression suite. CI configuration was committed but not exercised on GitHub. No public deployment or external organization identity provider was verified.

05 / ADOPTION BOUNDARIES

Fictional demo data only. No autonomous applicant ranking, job suitability scoring, AI-generated decisions or unvalidated prose. Scanned-image OCR, malware scanning, production TLS, organization SSO, encrypted backups, operational monitoring and enterprise retention policy require separate engineering and review.

SOURCE

github.com/keenosmith-del/deepseek-ai-workforce-planning